

Grassroots NGO Impact Measurement System

Moving from Activity-Tracking to Outcome Evaluation (Track 5.1)

The Core Problem: The "Activity vs. Outcome" Blindspot

- Most NGOs track operational activities (e.g., events held) instead of actual human outcomes (e.g., jobs retained).
- There is a missing translation layer between raw field data and executive decision-making.
- Without live "Cost-Per-Impact" visibility, donor capital is drained on inefficient, reactive budgeting.

Our Solution Framework: A 3-Tier Architecture

- **Tier 1 (Data Capture):** Standardized input form for field workers to eliminate human data entry errors.
- **Tier 2 (Processing):** Python/Pandas backend engine calculating KPIs based on the M&E Logical Framework.
- **Tier 3 (Visualization):** Streamlit web dashboard providing non-technical executives with a clean UI and live optimization algorithms.

The Prototype in Action: The Data Pipeline

Field_Worker_ID *

Enter Your ID here (format: FW_EmployeeNumber)

Your answer

District *

Select the district in which the course was held

Choose ▼

Beneficiaries_Served *

Enter the number of people who have completed this skill enhancement course

Your answer

Budget_Allocated_INR *

Your answer

Budget_Allocated_INR *

Enter below (in Indian Rupees) the budget allocated/spent for conducting this course and training the beneficiaries.

Your answer

Successful_Placements *

Enter the number of beneficiaries from this batch who got job placement.

Your answer

A copy of your responses will be emailed to aditi_s@me.iitr.ac.in.

Submit [Clear form](#)

Real-time data from the field instantly translates into executive KPIs.

Strategic Operations

- **The ONE number the ED should check?** Cost-Per-Impact. It instantly flags if a district is experiencing operational leakages.
- **Handling data quality issues:** Mandatory fields and dropdowns in the Form, combined with zero-division bypass logic in the Python backend.
- **The right reporting cadence:** Real-time data entry by field workers, but Weekly (Monday) dashboard reviews by the Executive Director for strategic budget reallocation.

Scalability & Projected Impact

- **Feasibility:** Built using free, open-source libraries (Python, Streamlit). Zero software licensing fees; loads in < 3 seconds.
- **Impact Projection:** Identifying performance asymmetry allows an NGO to reallocate just 15% of an upcoming budget from high-overhead zones to highly optimized operations.
- **The Result:** Projected +18.4% lift in capital efficiency, allowing the NGO to train hundreds of additional beneficiaries without needing new donor money.